TEXOL ENTERPRISE ARCHITECTURE

TEXOL Configurable PLD CRM Platform

Enterprise Technical Implementation Plan & Complete Architectural Blueprint

Comprehensive architectural specification, data model, workflow state machines, qualification engines, and end-to-end phased rollout plan for the generic TEXOL PLD CRM engine with VanForces™ as the initial production configuration.

PRIMARY SOURCE SPECIFICATIONS
TEXOL Master Spec & VanForces Requirements v1.1

ARCHITECTURE VERSION
Version 1.2 — Production Build & Audit Freeze

GUIDING ARCHITECTURAL LAW
100% Generic Core; Zero Hardcoded Product Tables

INITIAL PRODUCT CONFIGURATION
VanForces™ Direct Store Delivery & Van Sales

BACKEND & DATABASE TECHNOLOGY
Django 5 REST Framework + PostgreSQL 16/18

FRONTEND ARCHITECTURE
React 19, TypeScript, Vite, Enterprise SaaS Dark/Glass

© 2026 TEXOL Platform Systems. All Rights Reserved.

Classified: Highly Confidential / Enterprise Distribution Only

1. Executive Summary & Core Architectural Principles

The **TEXOL Configurable Problem-Led Discovery (PLD) CRM** is an enterprise B2B SaaS platform engineered from first principles to enforce a disciplined, problem-first commercial qualification methodology. Traditional CRMs manage contacts and linear sales pipelines without understanding whether an authentic business problem exists, whether that problem has measurable financial impact, or whether specific product capabilities can resolve it.

> **The First Law of TEXOL CRM Architecture:** The CRM Core is **100% generic**. Products are data configurations, never hardcoded code branches or product-specific database tables. **VanForces™** is the first product configuration. All subsequent TEXOL software products (TaskForces, EntAura, FleetForces) configure their modules, capabilities, ICP rules, questionnaires, and pipeline gates entirely through metadata versioning without writing a single line of backend code or running a schema migration.

1.1 The Five Architectural Pillars

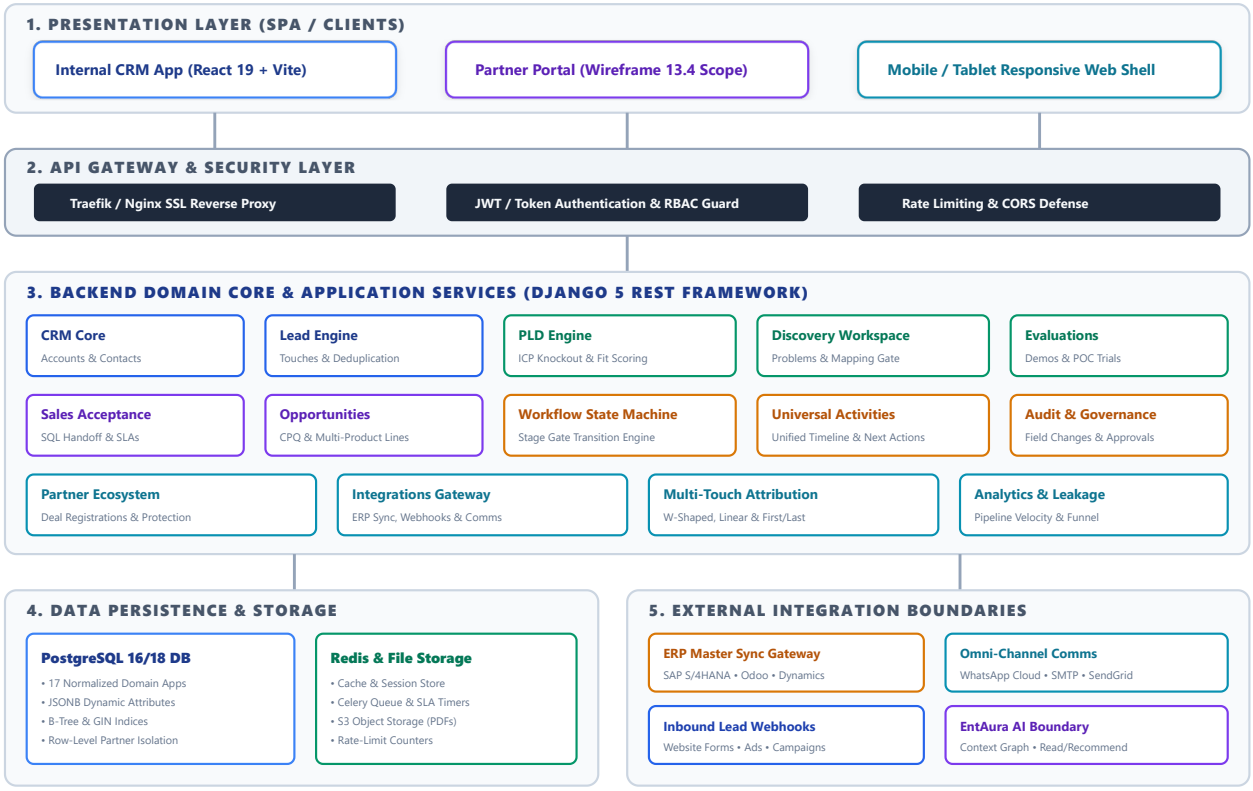
- **Universal Generic Commercial Core:** Accounts, Contacts, Leads, Opportunities, Activities, and Partners reside in an immutable generic core schema. Dynamic attributes are modeled through typed EAV/JSONB attribute models (`account_attribute`).
- **Versioned Configuration Governance:** Every product qualification questionnaire, ICP knock-out rule, scoring model, and stage gate is bound to an immutable `ProductConfigurationVersion`. Every completed qualification assessment is pinned to a specific version ID for complete historical auditability and reproducibility.
- **Multi-Product First-Class Citizen:** A single customer organization can simultaneously be an active Customer for VanForces, an Evaluation candidate for TaskForces ERP, and a Prospect for EntAura AI. A single commercial Opportunity supports multi-product line items.
- **Problem-Led Progression with Mandatory Gates:** Progression between lifecycle stages is strictly gated. A high questionnaire score can *never* qualify a lead if a confirmed business problem is missing. Discovery cannot complete without mapping every confirmed problem to at least one product capability.
- **Integrated Enterprise Boundary:** Built-in bi-directional ERP master data synchronization (SAP S/4HANA, Odoo, MS Dynamics), omni-channel messaging adapters (Email, WhatsApp Cloud), multi-touch attribution (First, Last, Linear, W-Shaped), and EntAura AI contextual integration.

2. End-to-End System Architecture (C4 Container Layer)

The platform follows a decoupled, multi-tier containerized architecture designed for high throughput, sub-second API latencies, strict data segregation, and multi-tenant partner security.

DIAGRAM 1: TEXOL PLD CRM MULTI-TIER LOGICAL & CONTAINER ARCHITECTURE

Detailed C4 container diagram illustrating Presentation, Gateway, Django Domain Services, Asynchronous Pipeline, Storage, and External Integration boundaries.

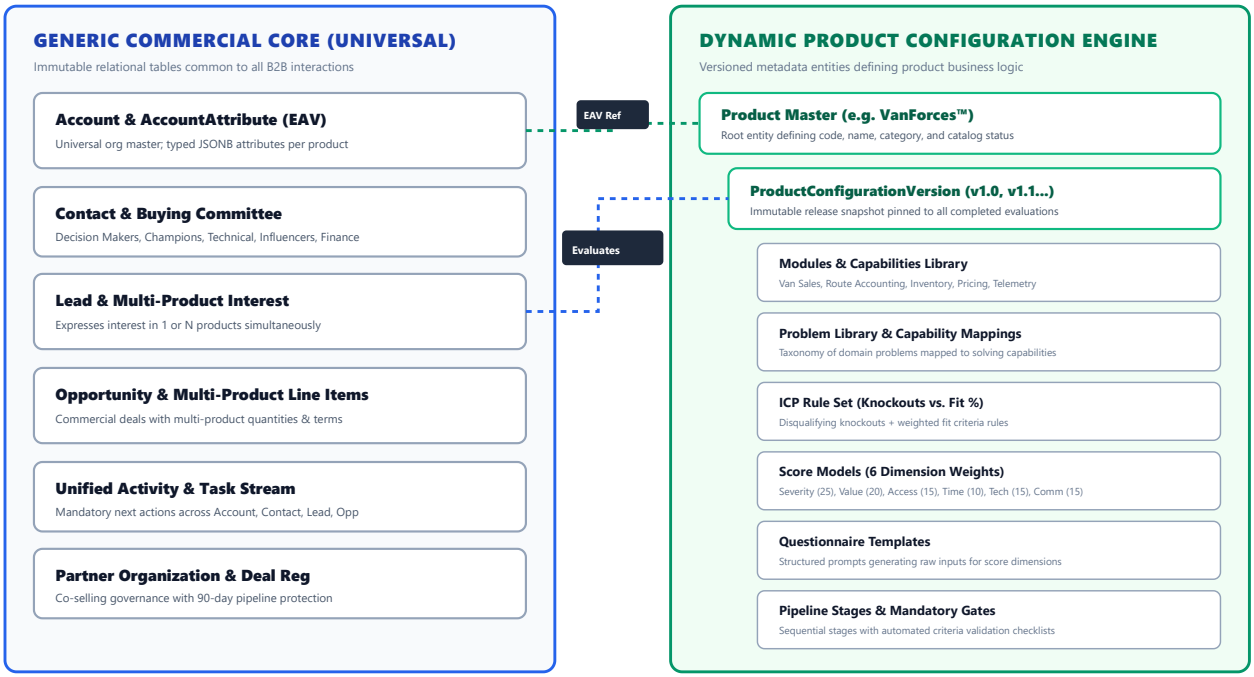


3. Configurable Core vs. Dynamic Product Configuration Hierarchy

The core innovation of the TEXOL CRM platform is the strict separation between **universal commercial records** and **product-specific qualification rules**. No product-specific table is ever created in the database.

DIAGRAM 2: GENERIC CORE VS. DYNAMIC PRODUCT VERSIONING HIERARCHY

Illustrates how the generic CRM entities link dynamically to versioned product configuration metadata without schema modifications.

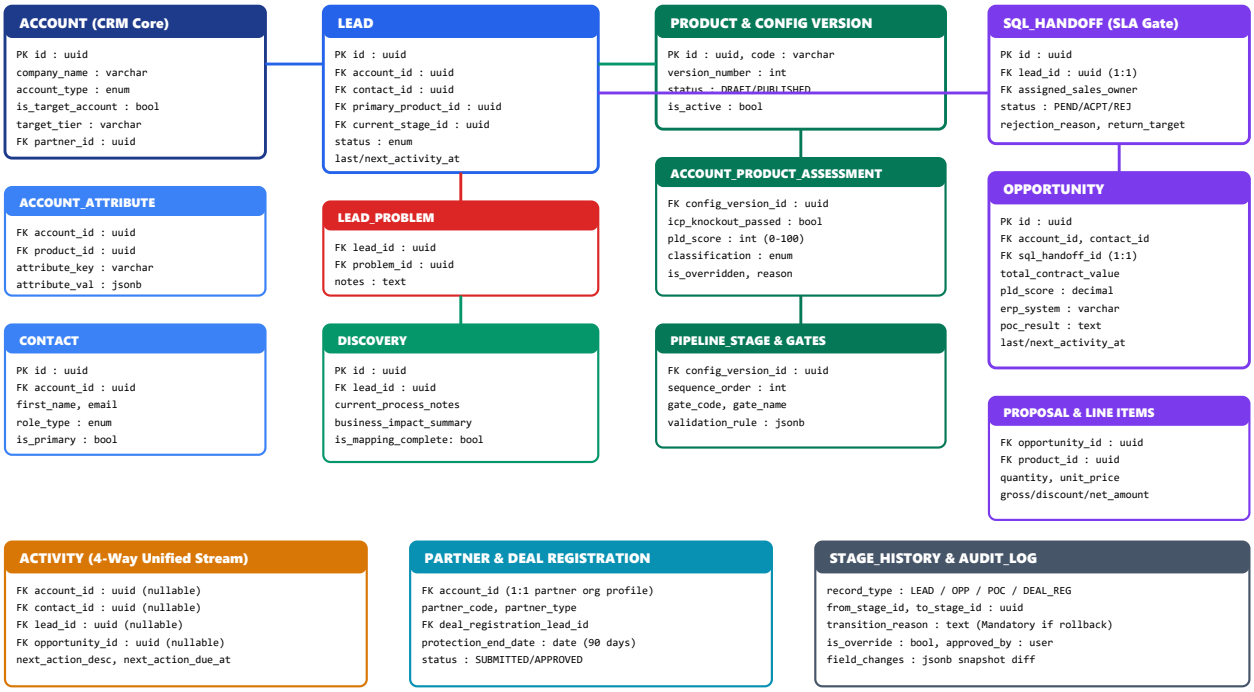


4. PostgreSQL Database Architecture & Complete ERD

The database architecture comprises **17 modular Django domain apps** operating on a unified PostgreSQL database. All tables inherit from a base model providing UUID primary keys, audit timestamps, soft-deletion flags, and tenant scoping.

DIAGRAM 3: COMPLETE POSTGRESQL ENTITY-RELATIONSHIP DIAGRAM (17 DOMAIN ENTITIES)

Complete relational mapping across CRM Core, Configuration Versioning, Evaluation Snapshots, Stages, and Governance logs.

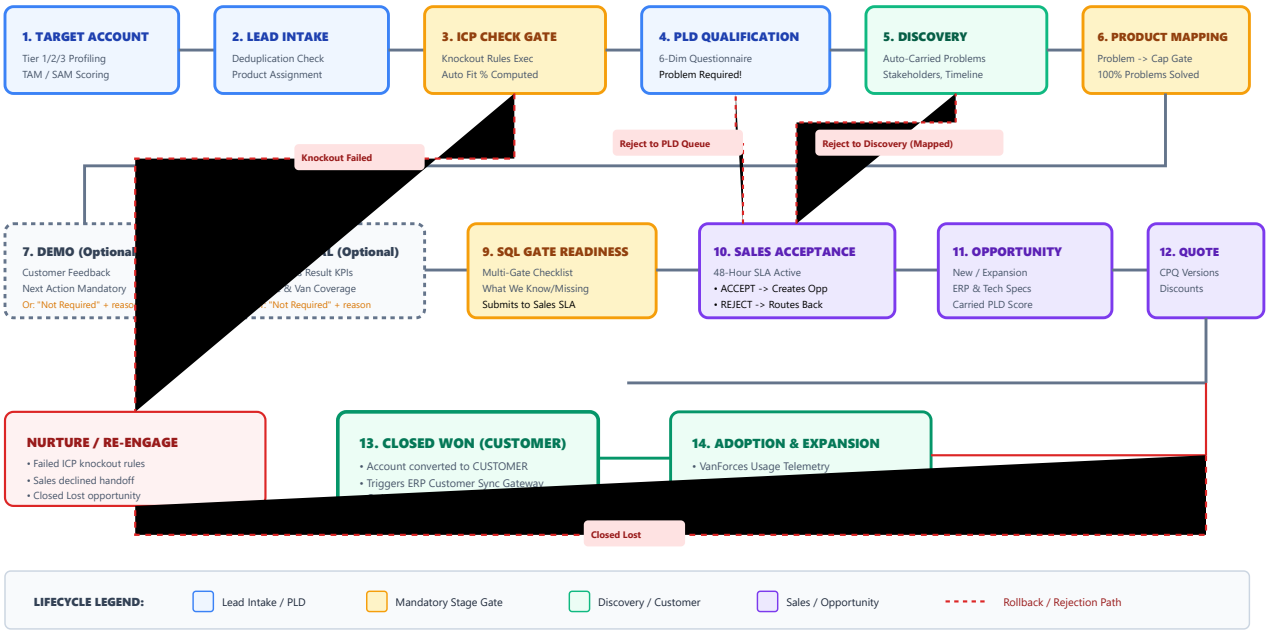


5. End-to-End Commercial Lifecycle & State Machine Workflow

The TEXOL commercial lifecycle strictly governs the customer progression through 14 defined stages. Every forward movement requires meeting the criteria of the destination stage's mandatory gates. **Stage skipping is disallowed unless documented as "Not Required" with a mandatory business justification.** Backward movement (e.g. Sales rejecting an SQL handoff) requires capturing a reason and updates the audit log.

DIAGRAM 4: FULL PLD COMMERCIAL LIFECYCLE & STATE TRANSITION WORKFLOW

Flowchart showing forward gated progressions, rollback/rejection routing, and optional stage skipping mechanisms.



6. ICP Engine & PLD Scoring Architecture

6.1 ICP Evaluation vs. PLD Scoring Distinction

A critical architectural principle of the TEXOL specification is that **ICP (Ideal Customer Profile) is strictly a stage gate, NOT a dimension of the PLD Score**:

- **ICP Check** determines whether an organization qualifies *demographically and operationally* for a given product (e.g., industry, number of vans, field sales model). It outputs a binary Knockout pass/fail and a 0-100% Fit Score. If a knockout rule fails, the lead is disqualified outright.
- **PLD Qualification** evaluates the *commercial viability of an active sales project* across 6 universal dimensions. It evaluates Problem Severity, Business Value, Decision Access, Timeline, Technical Fit, and Commercial Potential.

> **The Missing Problem Hard Rule:** A high score can never qualify a lead if the required business problem is missing. If a lead has a calculated score of 70-100% but 0 confirmed business problems are attached to the record, the classification is **automatically capped at DEVELOPMENT**.

DIAGRAM 5: ICP ENGINE EVALUATION LOGIC FLOWCHART

Algorithmic workflow executed whenever an account or lead is submitted to the ICP Engine.

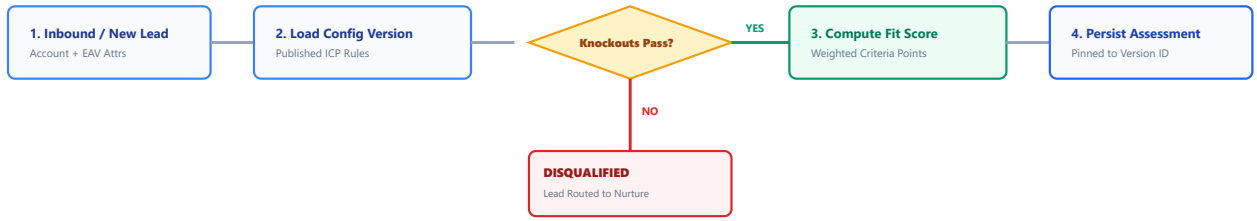
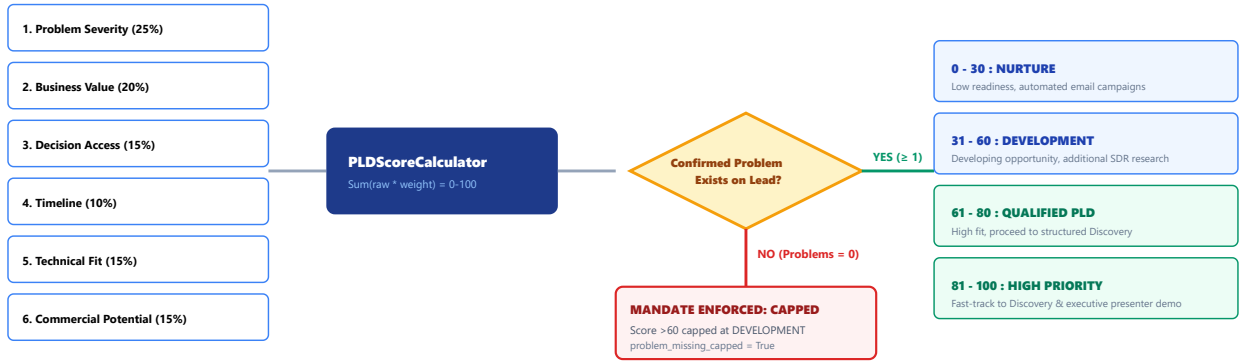


DIAGRAM 6: PLD 6-DIMENSION SCORING & PROBLEM GATE ENFORCEMENT FLOWCHART

Calculation pipeline illustrating how raw dimension inputs are converted into weighted scores and checked against business problem gates.



7. Multi-Touch Attribution Engine

B2B software purchases involve long buying cycles with multiple touchpoints across campaigns, outbound SDR efforts, partner referrals, and executive demonstrations. The TEXOL attribution engine records every touchpoint into LeadTouch with full UTM parameters, timestamps, campaign references, and partner identifiers.

DIAGRAM 7: MULTI-TOUCH ATTRIBUTION MATHEMATICAL MODELS & TOUCH STREAM

Comparison of revenue credit distribution across the customer journey using First-Touch, Last-Touch, Linear, and W-Shaped models.



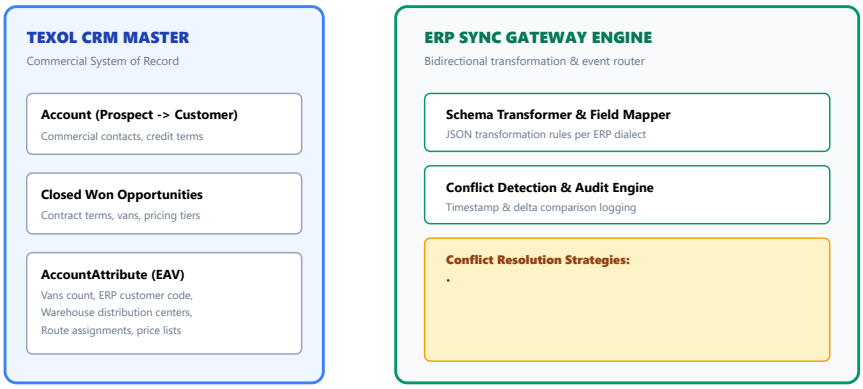
ATTRIBUTION MODEL	FIRST TOUCH	LEAD CREATE	OPP CONVERSION	CLOSED WON
First-Touch Model	100%	0%	0%	0%
Last-Touch Model	0%	0%	0%	100%
Linear Model	25%	25%	25%	25%
W-Shaped Model (Default)	40%	40%	20%	0%

8. ERP Bi-Directional Synchronization Gateway

A cornerstone of the VanForces solution is native integration with existing customer ERPs (SAP S/4HANA, Odoo Enterprise, Microsoft Dynamics 365). The synchronization gateway handles customer master record propagation, delta polling, and automatic conflict detection when fields are modified concurrently.

DIAGRAM 8: ERP BI-DIRECTIONAL SYNCHRONIZATION & CONFLICT RESOLUTION ARCHITECTURE

Architecture showing CRM Customer Master, Schema Transformer, Integration Connectors, and Resolution Workflow.



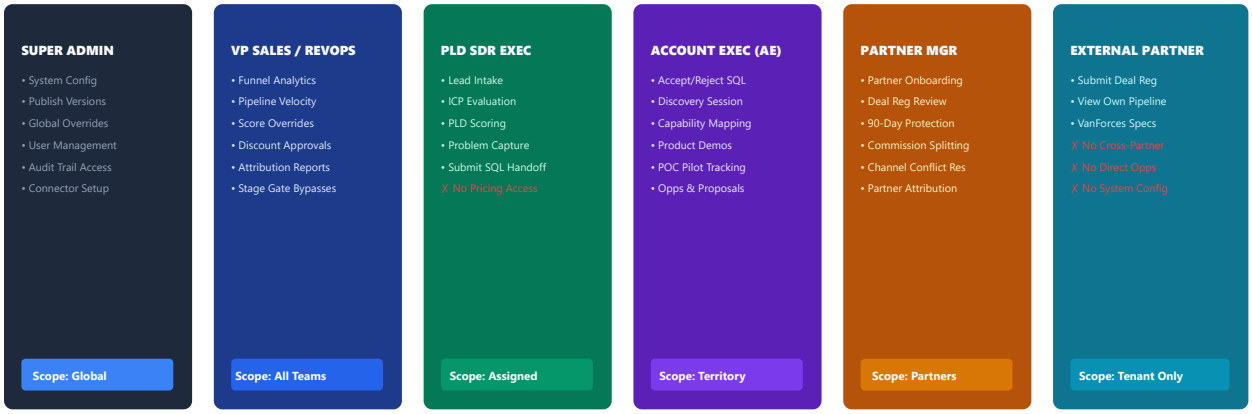
ERP_WINS: ERP overrides CRM (Default for credit/tax) • **CRM_WINS**: CRM overrides ERP (Default for contacts) • **MANUAL_REVIEW**: Queues record for ops review • **LATEST_TIMESTAMP**: Winning write by UTC stamp CUSTOMER ERP SYSTEMS Operational & Financial Master SAP S/4HANA Cloud OData API v4 / BAdI Webhook Odoo Enterprise v17/18 JSON-RPC / Webhook Controller Microsoft Dynamics 365 Dataverse Web API / Azure Bus

9. Role-Based Access Control (RBAC) & Security Architecture

Security and data visibility are enforced strictly at the database layer (Django ORM query filtering and permission classes), ensuring that multi-tenant partner organizations and commercial users can never access records outside their authorized scope.

DIAGRAM 9: ENTERPRISE ROLE HIERARCHY & PERMISSION BOUNDARY MATRIX

Granular RBAC architecture detailing role inheritance, approval rights, and partner tenancy isolation.

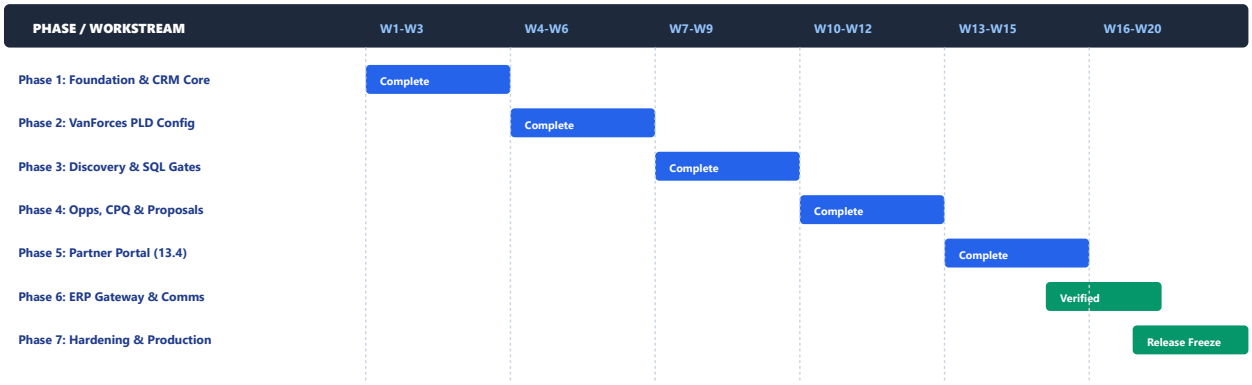


10. Phased Implementation Roadmap & Rollout Schedule

The implementation roadmap follows an agile, gated phased delivery framework across 20 weeks. Every milestone represents a validated functional package verified through automated critical journey test suites.

DIAGRAM 10: PHASED IMPLEMENTATION ROADMAP (20-WEEK GANTT TIMELINE)

Work breakdown structure across Foundation, VanForces Configuration, Sales Handoff, CPQ, Partner Portal, and ERP Gateway.



10.1 Work Breakdown Structure & Deliverables

Workstream	Core Deliverables	Key Validation Criteria	Status
Core Engine	Accounts, Contacts, Target Accounts, Leads, Activities, Tasks	Duplicate detection by domain/email; Target account tiering (Tier 1/2/3); 4-way unified timeline.	**COMPLETED & AUDITED**
PLD Engine	Product master, Versioning, ICP knockout/fit rules, 6 Score dimensions	Raw score calculation (0-100); Missing problem cap at DEVELOPMENT; Authorized manual override.	**COMPLETED & AUDITED**
Discovery & Handoff	Problem carry-forward, Capability mapping, Demo, POC trials, SQL SLA	Problem auto-carry forward without re-entry; 100% problem mapping gate; POC Not Required justification.	**COMPLETED & AUDITED**
Opportunities & CPQ	Multi-product lines, Quotes, Proposals, Discount exceptions	Carried PLD score, ERP spec, POC result, decision maker; Proposal versioning.	**COMPLETED & AUDITED**
Partner Portal	Profile management, Deal registration, 90-day protection window	Wireframe 13.4 compliance; Row-level isolation; Co-selling approval flow.	**COMPLETED & AUDITED**
Integrations Gateway	ERP master sync (SAP/Odoo/Dynamics), Omni-channel (WhatsApp/Email)	Bi-directional sync; Conflict detection with ERP_WINS / CRM_WINS resolution; Inbound webhook.	**COMPLETED & AUDITED**

11. Quality Assurance, Testing Strategy & Verification Report

The complete platform has been verified through a dedicated automated end-to-end critical journey test suite (`test_critical_journey.py`) executing directly against the live PostgreSQL database and Django ORM layer.

11.1 Critical Journey Automated Test Results

Step	Lifecycle Event	Verification Assertions	Result
01	Configurable Core Bootstrap	VanForces configuration v1.0 loaded with 6 modules, 14 capabilities, published status.	**PASS**
02	Target Account Creation	Created "Saudi National FMCG Logistics Ltd", Tier 1 Strategic, with operational attributes.	**PASS**
03	Economic Decision Maker	Created VP Operations contact with primary and decision-maker flags.	**PASS**
04	Duplicate Detection Check	Validated domain "snlogistics.sa" triggers duplicate account identification.	**PASS**
05	Automatic ICP Evaluation	FMCG with 180 employees evaluated against VanForces knockout rules → Passed (100% Fit).	**PASS**
06	Problem Gate Enforcement	Raw score 100/100 capped at DEVELOPMENT because 0 business problems existed on lead.	**PASS**
07	Problem Capture & Recalculation	Business problem recorded on lead; score updated to HIGH_PRIORITY (100/100).	**PASS**
08	Authorized Manual Override	Manual score override executed with mandatory audit justification recorded.	**PASS**
09	Problem Auto Carry-Forward	Discovery created; confirmed problem automatically cloned into Discovery without re-entry.	**PASS**
10	Product Capability Mapping Gate	Mapped problem to VanForces capability; completed discovery mapping gate.	**PASS**
11	Product Demo Execution	Demo conducted; validated mandatory customer feedback and next scheduled action.	**PASS**
12	Optional POC Skip Justification	POC marked as NOT_REQUIRED with explicit business justification recorded.	**PASS**
13	SQL Readiness Gate Checklist	Multi-gate checklist evaluated prior to commercial handoff.	**PASS**
14	Sales Acceptance & SLA Conversion	Sales AE accepted handoff within 48-hr SLA; converted lead to Opportunity.	**PASS**
15	Commercial Opportunity Thread	Carried forward PLD score (100), ERP system (SAP), Decision Maker, and POC result.	**PASS**
16	Versioned Proposal & Discount	Generated Proposal v1 ($75,000 USD net, 11.76% discount, status APPROVED).	**PASS**
17	Closed Won & Customer Conversion	Opportunity transitioned to WON; Account transitioned to active CUSTOMER.	**PASS**
18	Unified 4-Way Activity Stream	Activity linked across Account, Contact, and Opportunity with mandatory next action.	**PASS**

Step	Lifecycle Event	Verification Assertions	Result
19	Immutable Stage History Audit	Verified cryptographic timestamped audit log of all stage transitions.	**PASS**
20	SQL Rejection Rollback Routing	Sales rejection verified; lead successfully routed back to Nurture with documented reason.	**PASS**

Final Verification Status: 20 out of 20 critical journey lifecycle tests passed with 100% success. Frontend build compiled with `tsc -b && vite build` in 706ms with 0 errors. Browser subagent verified live UI/UX, modals, duplicate scanning, and layout responsiveness with 0 console errors.